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Modified Mann-Kendall trend test for hydrological time series under the scaling hypothesis and its application

Zichen Hu^a, Shuguang Liu^{a,b}, Guihui Zhong^a, Hejuan Lin^c and Zhengzheng Zhou^a

^aDepartment of Hydraulic Engineering, Tongji University, Shanghai, China; ^bKey Laboratory of Yangtze River Water Environment, Ministry of Education, Shanghai, China; ^cBureau of Hydrology (Information Center), Taihu Basin Authority of Ministry of Water Resources, Shanghai, China

ABSTRACT

The existence of persistence contaminates the results of the Mann-Kendall (MK) trend test. This paper proposes an improved MK test for hydrological data with both positive and negative persistence. To this end, the variance and distribution patterns of S were modified under the scaling hypothesis. Different estimators of the scaling coefficient were compared. According to the results, the generalized normal distribution is the best approximation to fit the exact distribution of S with both positive and negative persistence. The centred detrending moving average was demonstrated to be the best estimator of the scaling coefficient for series with a trend. The improved MK test was proven to be more advantageous than other trend detection methods for the synthetic time series and observed precipitation data obtained from nine hydrometric stations located in China. With this modification, the application of the MK test can be extended to long-term persistent data.

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1 Introduction

The global climate change over the past several decades is a major concern because of the increasing frequency of extreme climate events such as extreme precipitation and temperature. The changing climate invalidates the stationarity assumption, affecting traditional approaches of frequency analyses and the definitions of return period, risk and reliability (Salas *et al.* 2018). The main approach used for the nonstationarity detection of hydrometeorological variables is the trend analysis. Many studies have been conducted on the trend detection of hydrological observations (Gautam and Acharya 2012, Han *et al.* 2014, Rashid *et al.* 2015, Yilmaz and Perera 2015, Ledvinka and Lamacova 2015, Wang *et al.* 2015, Wang 2016, Ishak and Rahman 2019). The mainly used methods can be broadly divided into two types. The first type is trend slope estimation, e.g. regression analysis and Sen's slope estimation (Sen 1968), which are representative parametric and non-parametric methods, respectively. The other type is a trend significance test, which tells whether the trend in a target time series is significant at a preassigned significance level. The widely used methods are the Mann-Kendall (MK) trend test (Mann 1945, Kendall 1975) and Spearman's rho (RS) trend test (Lehmann and D'Abrera 1975, Sneyers 1990). A newly developed method is the innovative trend analysis (ITA, Sen 2012). The present study focuses on trend significance test methods. Yue *et al.* (2002) demonstrated that the MK and SR tests give similar results for the monotonic trend detection of hydrological time series. Therefore, only the MK test is analyzed in this study.

The MK trend test is one of the most widely used trend detection methods. Being rank-based, the MK trend test is unaffected by the actual distribution of the data even when significantly skewed with some outliers, making it quite suitable for hydrological data. Over the past several decades, various versions of the adjusted MK test have been developed and applied for different targets. For example, Hajani *et al.* (2017) applied the regional MK test (Hirsch *et al.* 1982, Hirsch and Slack 1984, Douglas *et al.* 2000) to detect the trend in rainfall intensity on the regional scale and the partial MK test (El-Shaarawi 1993, Libiseller and Grimvall 2002) to evaluate the impact of trends on climate index data in NSW, Australia. One of the basic assumptions of the original MK test is the independence of observations. Cox and Stuart (1955) and Koutsoyiannis (2003) stated that if the MK test is used without considering the persistence, the trend might be incorrect, which has been the case for most types of hydrological data. Some countermeasures were proposed to eliminate this influence, in terms of the choice of persistence model used, estimation of persistence coefficient, and modification of the MK algorithm.

Since the discovery of the Hurst phenomenon during an investigation on the discharge of Nile River (Hurst 1951), the persistence of hydrological data has been considered a long-term persistence (LTP), which describes a time series as a combination of multi-scale stochastic fluctuations. Hurst (1951) proposed the coefficient H , known as the Hurst coefficient or scaling coefficient, to quantify the persistence level of LTP series. A series is said to be independent when $H = 0.5$, while it is positively (negatively) persistent when $H > 0.5$.

($H < 0.5$). A persistence model based on the coefficient H , namely the scaling model, was developed to numerically describe series with LTP; “data exhibit scale invariance at any scale greater than annual” (Koutsoyiannis 2003). The scaling model was considered more reasonable in describing the persistence of hydrological data than short-term models, since LTP-like patterns, such as the geological history of ice ages and sea-level changes, are widely observed in long records (Cohn and Lins 2005).

Since the Hurst phenomenon is widely detected in hydrological observations, some algorithms have been proposed to estimate H , such as the rescaled range (R/S) analysis (Mandelbrot and Wallis 1968), Higuchi method (Higuchi 1988), and periodogram (Taqqu *et al.* 1995). The widely used estimator of H is the R/S analysis, owing to its convenient calculation and low time complexity. However, applying the R/S method requires the absence of serial trends; otherwise, H would be overestimated (Yue and Wang 2004). To eliminate the influence of such trends, Hamed and Rao (1998) applied the Sen’s non-parametric trend estimation (Sen 1968) to remove the apparent trend before estimating its persistence. Although this pretreatment method has been widely used in practice, its effectiveness remains in question, because the trend is treated as a linear deterministic component despite containing randomness. Arianos and Carbone (2007) presented a time-domain method, namely the centred detrending moving average (CDMA), which is free from the influence of trends if the parameters are correctly set (Shao *et al.* 2015).

Methods for modifying the trend tests applied to persistent data can be divided into two types. In the first type, the correlation in the time series is eliminated, e.g. the pre-whitening (PW) method first proposed by Von Storch (1995). In this method, the serial correlation is estimated and removed before the trend test. The technique was then developed into trend-free pre-whitening (TFPW) for both positively (Yue and Wang 2002) and negatively (Rivard and Vigneault 2009) correlated series and has been widely applied to hydrological time series (Abdul Aziz and Brun 2006, Gautam and Acharya 2012, Tabari *et al.* 2012, Ledvinka and Lamacova 2015). However, the performance of the pre-whitening technique in correctly eliminating the serial correlation remains unclear (Sang *et al.* 2014, Wang *et al.* 2015). Razavi and Vogel (2018) found “loss of information” during pre-whitening, which eventually led to underestimation of extreme conditions. Sen (2016) proposed the over-whitening (OW) approach, which adds white noise with zero mean and specific standard deviation to eliminate the serial correlation. In the other type of methods, the distribution of the statistic due to persistence is modified. A common technique is the effective sample size (ESS) proposed by Matalas and Langbein (1962), who documented that a time series with serial correlation contains less information than its actual serial size. Lettenmaier (1976) used the ESS to correct the power of the t-test and Spearman’s rho test. Yue and Wang (2004) demonstrated that the ESS approach (BHMLL-ESS) can effectively eliminate the effect of serial correlation in the MK test with no significant trend. Hamed and Rao (1998) introduced a modified MK trend test

for arbitrary autocorrelation structures. Matalas and Sankarasubramanian (2003) determined the variance of the regression coefficient using the correlation models of independence, AR (1), ARMA (1,1), and FGN (called RA method in this paper). Considering Hamed (2008, 2009) modified the variance and distribution forms of S under a scaling hypothesis and proposed the beta distribution, instead of the normal distribution, to fit the S distribution of a moderate-length series with positive LTP. However, no reasonable approximation of the S distribution for LTP data has been studied for the entire range of H , i.e. for both positive and negative persistent data, despite widely detecting negative persistence in hydrological data (Rivard and Vigneault 2009).

This study aims to modify the MK trend test based on the scaling model with the entire domain of H . In this paper, the MK method is first introduced, and the influence of persistence is illustrated theoretically. The scaling model is then introduced, and its impact on the original MK test is evaluated through a Monte Carlo simulation. A theoretical and empirical expression of S variance under the scaling hypothesis is proposed. The generalized normal (GN) distribution with a continuous shape parameter k is proven to better approximate S than the normal and beta distributions. The modified MK test outperforms other widely used methods, as demonstrated by Monte Carlo simulation. Thereafter, to give a better estimation of H , the widely used estimators are compared in terms of their unbiasedness and uncertainty. Thus, an improved MK trend test procedure, including estimation of H , is proposed. Finally, the modified MK procedure is applied to precipitation data obtained from nine stations located in China to re-assess the significance in their trends as a case study.

2 Mann-Kendall trend test

The MK trend test is a rank-based significance test, which judges the significance of trends by testing whether S of the target time series falls in the confidence interval of the null hypothesis of the preassigned significance level. Being non-parametric, the MK test is applicable to both Gaussian and non-Gaussian data. Here, Gaussian data were taken to derive the mean and variance of the S distribution. The application to non-Gaussian data is discussed in Section 7. For a time series $X = \{x_1, x_2, x_3, \dots, x_N\}$, S is defined as

$$S = \sum_{i < j} a_{ij} \quad (1)$$

where a_{ij} denotes the rank relationship between x_i and x_j

$$a_{ij} = \text{sgn}(x_j - x_i) = \begin{cases} 1 & x_i < x_j \\ 0 & x_i = x_j \\ -1 & x_i > x_j \end{cases} \quad (2)$$

The positive (negative) value of S indicates increasing (decreasing) trend of the series versus time. Under the assumption that the data are independent and identically distributed, S tends to normality when N is large enough (Kendall and Maurice 1955).

By setting the independence as a basic assumption at first, we discuss the cases of persistent data later in this section. Under the null hypothesis of stationarity, x_i and x_j both obey $N(\mu, \sigma^2)$, and $x_j - x_i \sim N(0, 2\sigma^2)$, which is the mean of the S distribution, is given by

$$\begin{aligned} E(S) &= E\left(\sum_{i < j} a_{ij}\right) = \sum_{i < j} E(a_{ij}) \\ &= \sum_{i < j} [P(x_j - x_i > 0) - P(x_j - x_i < 0)] \end{aligned} \quad (3)$$

Because $x_j - x_i$ is symmetrically distributed, $P(x_j - x_i > 0) = P(x_j - x_i < 0)$. Equation (3) can be written as

$$E(S) = 0 \quad (4)$$

With the mean of zero, the variance of the S distribution is given by

$$\begin{aligned} V_0(S) &= E(S^2) - E^2(S) = E\left[\left(\sum_{i < j} a_{ij}\right)^2\right] \\ &= E\left(\sum_{i < j, k < l} a_{ij}a_{kl}\right) \end{aligned} \quad (5)$$

The RHS of Equation (5) contains $N^2(N-1)^2/4$ terms, since it should traverse all the suffixes of the whole series, which could be sorted into three items on the basis of the suffixes:

$$\begin{aligned} V_0(S) &= C_N^2 E(a_{ij}^2) + 2C_N^3 [E(a_{ij}a_{ik}) + E(a_{ij}a_{jk}) + E(a_{ik}a_{jk})] \\ &\quad + C_N^4 C_4^2 E(a_{ij}a_{kl}) \end{aligned} \quad (6)$$

where C_p^q denotes the number of combinations selecting q in p elements regardless of their orders. Because a_{ij} and a_{kl} are independent, $E(a_{ij}a_{kl}) = E(a_{ij})E(a_{kl}) = 0$. Equation (6) can be rewritten as

$$V_0(S) = \frac{N(N-1)}{2} + 2C_N^3 [E(a_{ij}a_{ik}) + E(a_{ij}a_{jk}) + E(a_{ik}a_{jk})] \quad (7)$$

According to Kendall and Stuart (1976), for a bivariate normal distribution, the following relationship holds:

$$E(a_{ij}a_{kl}) = P(a_{ij}a_{kl} = 1) - P(a_{ij}a_{kl} = -1) = \frac{2}{\pi} \sin^{-1}(\rho_{YZ}) \quad (8)$$

where $Y = x_j - x_i$, $Z = x_l - x_k$. Since $Y, Z \sim N(0, 2\sigma^2)$, $E(a_{ij}a_{ik}) = E(a_{ij}a_{jk}) = \frac{1}{2}$ and $E(a_{ik}a_{jk}) = -\frac{1}{2}$. Therefore, Equation (7) can be transformed to

$$\begin{aligned} V_0(S) &= \frac{N(N-1)}{2} + \frac{N(N-1)(N-2)}{9} \\ &= \frac{N(N+1)(2N+5)}{18} \end{aligned} \quad (9)$$

which is the commonly used expression for the variance of S distribution when using the MK test. The standardized statistic Z is given by

$$Z = \begin{cases} \frac{S-1}{\sqrt{V(S)}} & S > 0 \\ 0 & S = 0 \\ \frac{S+1}{\sqrt{V(S)}} & S < 0 \end{cases} \quad (10)$$

which follows the standard normal distribution. The significance of the trend can be tested by comparing the value of Z with the critical value α of the desired significance level. This method based on the independence assumption is called the MKO (the original MK trend test) in this paper.

When tied ranks exist, the variance of S becomes

$$V^*(S) = [N(N-1)(2N+5)]/18 - \sum_{j=1}^m t_j(t_j-1)(2t_j+5)/18 \quad (11)$$

where m is the number of groups of ties, and t_j is the number of tied observations of each group. However, since this research focuses on the influence of persistence, the impact of tied ranks that accompanies persistence is not discussed.

When persistence exists, $x_j - x_i \sim N(0, 2\sigma^2 - 2\sigma^2\rho_{ij})$ is still symmetric, and the expectation of S remains zero, where ρ_{ij} is the correlation between x_i and x_j . As a_{ij} is no longer independent of a_{kl} , Equation (9) for the variance of S is no longer satisfied. Equation (5) can be transformed into a more general form according to Equation (8) as follows:

$$\begin{aligned} V(S) &= \sum_{i < j, k < l} \frac{2}{\pi} \sin^{-1} \left[\frac{\text{cov}(x_j - x_i, x_l - x_k)}{\sqrt{V(x_j - x_i)V(x_l - x_k)}} \right] \\ &= \sum_{i < j, k < l} \frac{2}{\pi} \sin^{-1} \left[\frac{\rho_{jl} + \rho_{ik} - \rho_{jk} - \rho_{il}}{2\sqrt{(1-\rho_{ij})(1-\rho_{kl})}} \right] \end{aligned} \quad (12)$$

As the variance depends on the correlation between the observations, the MK trend test should be justified by the persistence level.

3 Influence of scaling hypothesis

3.1 Description of the scaling hypothesis

Since the concept of the Hurst phenomenon was put forward, LTP has been detected many hydrological observations. The scaling hypothesis was applied to reproduce the Hurst phenomenon (Bras and Rodríguez-Iturbe 1993, p. 221). We consider a time series X_t , with $t = 1, 2, 3, \dots, N$ denoting the discrete time. Data under the scaling hypothesis exhibit scale invariance at any time scale greater than annual (Koutsoyiannis 2003).

$$\left(Z_i^{(k)} - k\mu\right) \stackrel{d}{=} \left(\frac{k}{l}\right)^H \left(Z_j^{(l)} - l\mu\right) \quad (13)$$

where $\stackrel{d}{=}$ means equality in distribution, μ is the expectation of X_t , H is the scaling coefficient between 0 and 1, and k and l are positive integers representing two different time scales greater than the basic one. $Z_i^{(k)}$ is the aggregated stochastic process on time scale k denoted as

$$Z_i^{(k)} := \sum_{l=(i-1)k+1}^{ik} X_l \quad (14)$$

and $Z_j^{(l)}$ is the aggregated process on scale l . X_t following such a hypothesis could be summarized as an aggregation of multi-scale stochastic fluctuations. When X_t is normally distributed at the same time, it is known as fractional Gaussian noise (FGN) (Mandelbrot 1965).

Under the scaling hypothesis, data tend to be independent when H close to 0.5; otherwise, it presents persistence. Unlike the persistence pattern of STP models, the autocorrelation coefficient ρ_k of a scaling series decreases much slower with increasing lag k , owing to its multiple scale components. Figure 1 shows the difference between the two patterns of persistence models. The simulated values are obtained from 5000 iterations of Monte Carlo simulations with a series length of 5000 for each group, which are large enough to eliminate the influence of randomness. In this study, synthetic LTP series with preassigned H values were generated by the function *SimulateFGN* in the package *FGN* in R studio based on the algorithm proposed by McLeod *et al.* (2007). The calculated values follow the autocorrelation functions of the AR (1) model (Equation (15)) and scaling model (Equation (16), Koutsoyiannis 2003), respectively, where ρ_k denotes the autocorrelation with time lag k .

$$Z = \begin{cases} 1 & k = 0 \\ \rho_1^k & k > 0 \end{cases} \quad (15)$$

$$\rho_k = \frac{1}{2} (|k+1|^{2H} - 2|k|^{2H} + |k-1|^{2H}) \quad (16)$$

3.2 Influence of scaling on the original MK test

As demonstrated in Section 2, the existence of LTP impacts the distribution of S under the null hypothesis. According to Cohn and Lins (2005), the inconsistency between the null hypothesis and the natural system can lead to some wrong rejections. Monte Carlo simulations were adopted to test the impact of LTP on the MK results. Five thousand synthetic FGN series were generated for each group with different series length $N = 2^5, 2^6, \dots, 2^{10}$ and scaling coefficient

$H = 0.1 \text{ (0.1) } 0.9$. A linear trend of the mean $\beta = 0.000 (\pm 0.001) \pm 0.009$ was superimposed onto each series.

Figure 2 shows the rejection rate of the two-sided MK test with a 95% confidence level. The results of data with decreasing trends are omitted because they perform similarly with positive ones. For identically distributed data, the existence of trend increases the rejection rate of the null hypothesis, and the effect is more pronounced with larger N . This shows the conservation of the MK test for cases of small series length since it cannot clearly tell slight trends from randomness. For data with fixed β and N , a positive LTP ($H > 0.5$) increases the rejection rate while a negative persistence ($H < 0.5$) decreases it in most cases. The impact decreases with increasing β and N . In extreme cases ($\beta > 0.006$ and $N > 128$), a positive persistence decreases the rejection rate instead.

As noted, the LTP impacts the MK results by changing the variance of S . With a combination of Equations (12) and (16), the S variance of the scaling time series can be theoretically calculated. To recognize the influence of H more intuitively, the variance variation factor (VVF) is introduced, which represents the reduction in information due to serial correlation. The expression of $V(S)$ is:

$$V(S) = \text{VVF} \cdot V_0(S) \quad (17)$$

where $V_0(S)$ is the variance of the independent data in Equation (9). Figure 3(a) shows the theoretical VVF values at different LTP levels versus N in the form of a log-log plot. VVF equals 1 in the absence of persistence, and it increases (decreases) with increasing (decreasing) H and N .

To further demonstrate how scaling impacts the MK test besides variance change, the frequency distributions of S of different samples were compared with the S distribution calculated using the MK functions. Figure 4 shows a comparison with $N = 150$ and $\beta = 0$ as an example. “Exact” indicates the frequency distribution of S of the sample, and “Normal” indicates the probability density curve of the normal distribution with mean zero and variance calculated using Equation (10), while the curve “Norm*” obeys the normal distribution with the theoretical variance of the scaling data. The existence of positive persistence increases the variance, thus decreasing the confidence level of the same critical value, and vice versa. Moreover, “Norm*” in

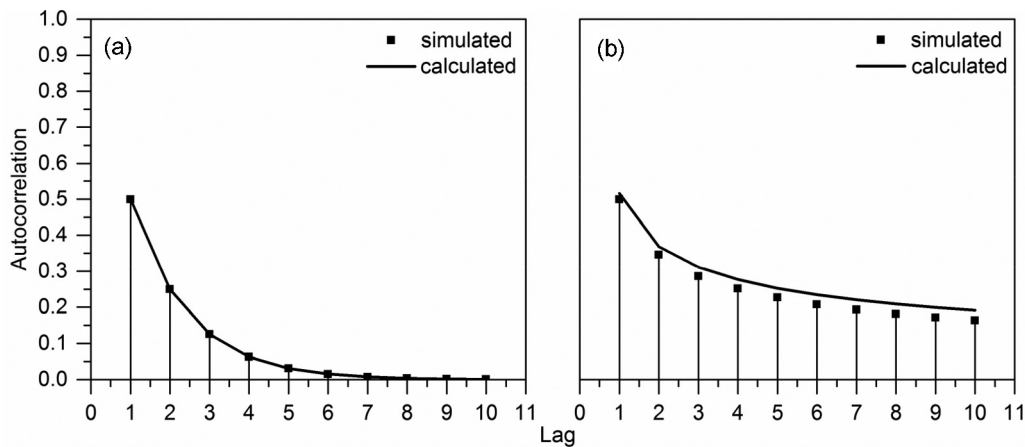


Figure 1. Comparison of autocorrelation versus lag between (a) AR (1) model and (b) FGN model versus lag.

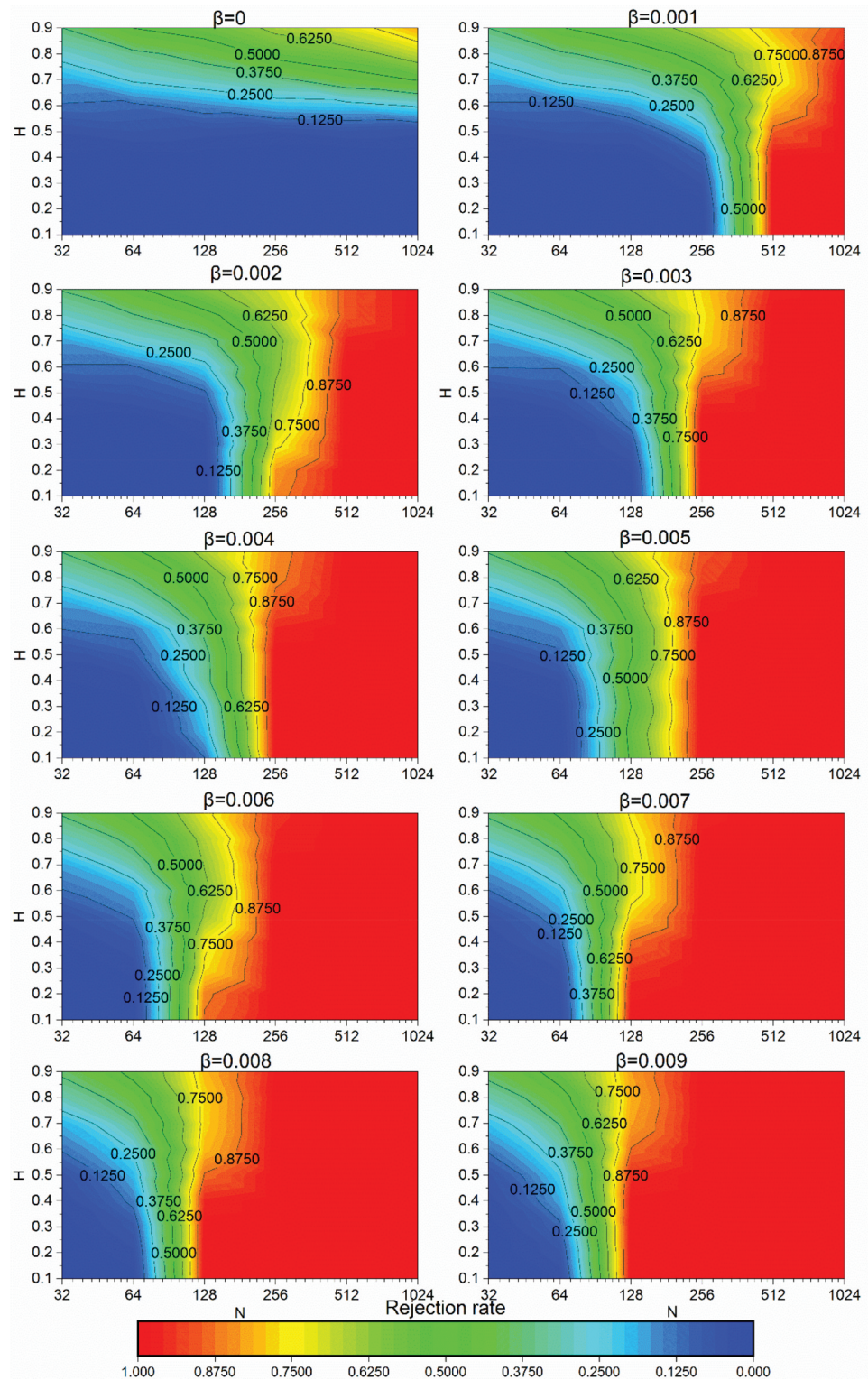


Figure 2. Rejection rate of the null hypothesis of the original MK trend test applied to FGN series of different H and N with varying magnitudes of linear trends, with 5000 time series iterations for each group.

Fig. 4(b) fits well with “Exact”, which is not the case in Fig. 4(a) and (c). This indicates that when the series presents LTP, the shape of the exact frequency density distribution of S as well as its variance changes. With positive (negative) persistence, the center and tail become thinner (heavier), and the other parts get heavier (thinner) (Fig. 4(a) and (c)). Both the changes in the variance and shape of the distribution misled the results of the MKO.

4 Modified Mann-Kendall trend test

4.1 Modification of the variance of S

As stated in Section 3.2, the S variance of a scaling series can be calculated using Equations (12) and (16). However, the theoretical calculation is impractical for mass data because of its high computational cost. Hence, it is

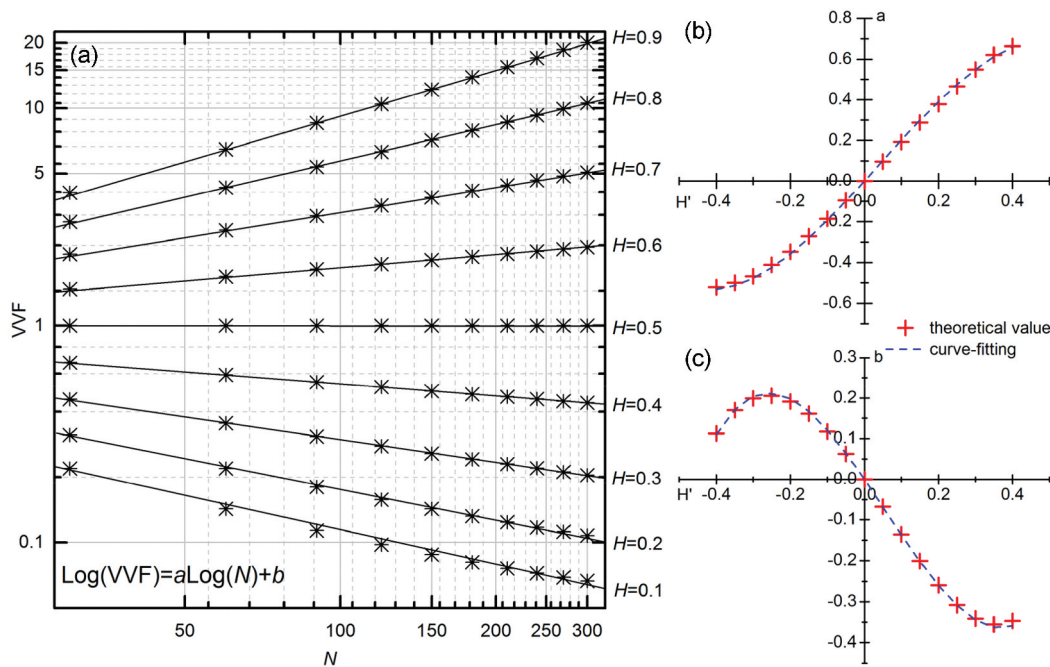


Figure 3. (a) Log-Log plot of the theoretical variance variation factor (VVF) of different N and H and the results of linear regression. (b) and (c) Curve fitting results of the coefficients a and b in the regression function, respectively.

meaningful to develop an empirical expression for the variance with both acceptable accuracy and low computational complexity. Hamed and Rao (1998) proposed an empirical expression for $V(S)$ based on the rank correlation for serial correlated data. Although this method reduces the computational complexity, the accuracy is only ensured for data obeying STP models. Given that the VVF presents a largely linear trend versus N with constant H in the log-log plot, shown in Fig. 3(a), the following function is adopted to fit the relationship.

$$\log(\text{VVF}) = a\log N + b \quad (18)$$

where the parameters a and b are functions of H . Figure 3(b) and (c) respectively shows the relationship between a , b , and H' , where H' is defined as $H - 0.5$ to tell positive and negative persistence zones apart. The empirical expressions of a and b can be obtained from the Levenberg–Marquardt algorithm as follows.

$$\begin{aligned} a &= -\frac{2H'}{5} (8H'^2 - H' - 5) \\ b &= \frac{H'}{9} (42H'^2 - 7H' - 12) \end{aligned} \quad (19)$$

Figure 5 shows the relative error of VVF fitting. The maximum relative error is 6.52% occurring when $N = 300$ and $H = 0.1$, while the errors in most other cases are acceptable (under 5%), proving the accuracy of the approximation. Moreover, the empirical expression of the variance decreases the time complexity from $O(N^4)$ to $O(1)$, making it applicable to hydrological data to reduce computing cost.

4.2 Modification of the distribution pattern of S

In MKO, the normal distribution is used to describe the S distribution of independent data. According to Hamed (2009), it remains a reasonable approximation for persistent data with large N . However, he proposed the Beta distribution for moderate-length series with positive persistence due to the shape change in the distribution of S . However, as previously noted, a negative LTP shows an inverse impact on the shape of the distribution of S , in which case the beta distribution cannot fit it well globally. To adapt to this feature, a continuous shape parameter k is added to the distribution function in this study. The generalized normal (GN) distribution is used here as follows:

$$f(x) = c_1 \sigma^{-1} \exp(-|c_0 z|^k) \quad (20)$$

$$F(x) = \begin{cases} 0.5 \left(1 + \frac{\Gamma_{|c_0 z|^k(1/k)}}{\Gamma(1/k)} \right) & x \geq \mu \\ 0.5 \left(1 - \frac{\Gamma_{|c_0 z|^k(1/k)}}{\Gamma(1/k)} \right) & x < \mu \end{cases} \quad (21)$$

where $c_0 = \left(\frac{\Gamma(3/k)}{\Gamma(1/k)} \right)$, $c_1 = \frac{k c_0}{2\Gamma(\frac{1}{k})}$, and $z = \frac{x - \mu}{\sigma}$. σ is the standard deviation, and k is a continuous shape parameter. When $k = 2$, the GN distribution is equivalent to the normal distribution; otherwise, the distribution deforms with increasing or decreasing k . Therefore, the GN distribution theoretically performs as well as the normal distribution at least when fitting “Exact” density distribution of statistic S . For comparison with the beta distribution, the two alternatives were applied to 5000 time series of each group using the maximum likelihood estimation. Table 1 lists the results of the K-S test statistic. The GN distribution passed the test of significance level $\alpha = 0.1$ in most cases with the exception of $H = 0.1$ and 0.2 with $N = 30$. For the beta distribution, the null

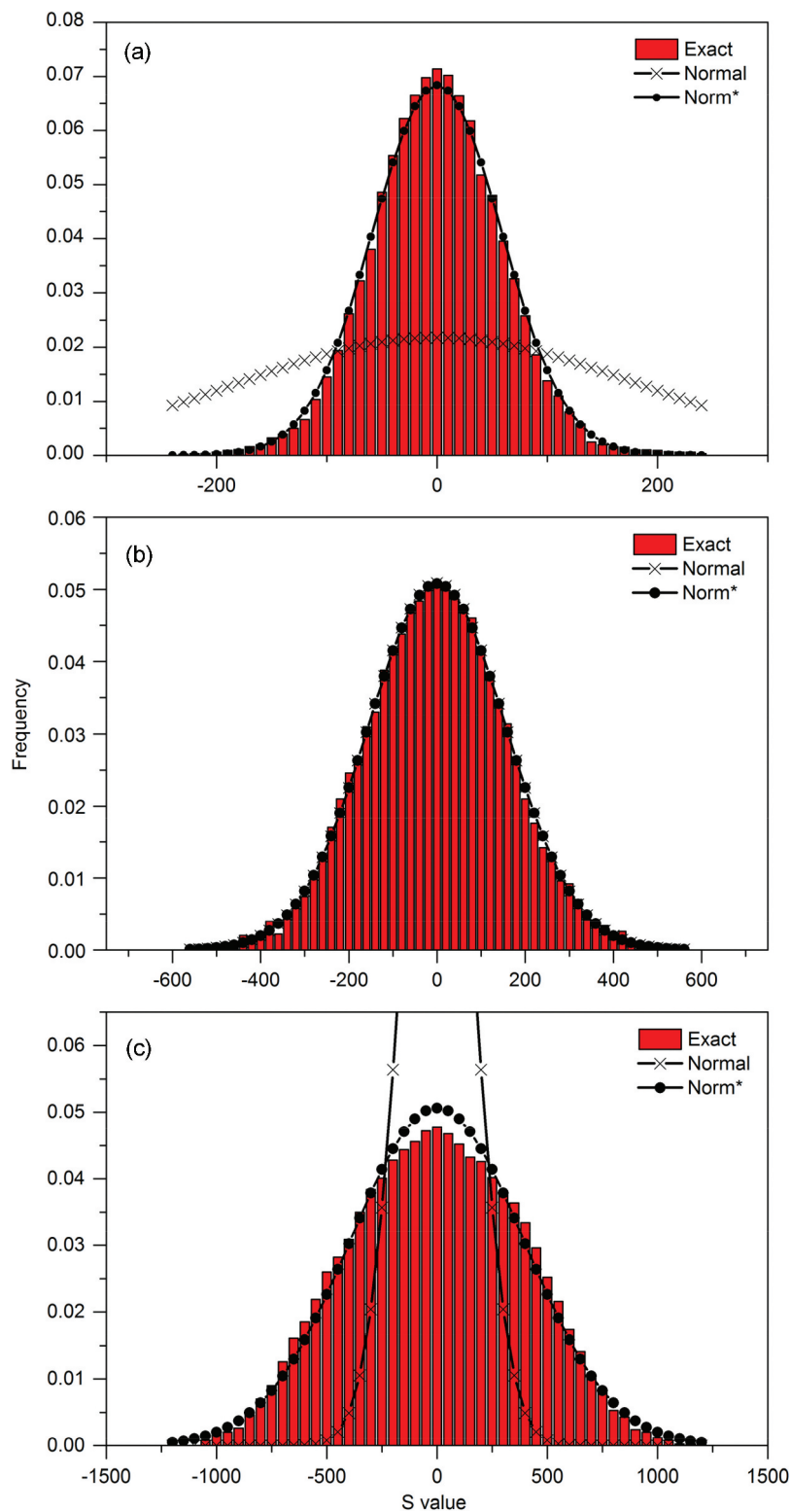


Figure 4. Comparison between the simulated frequency distribution of statistic S and the normal distribution with $N = 150$, $\beta = 0$ and (a) $H = 0.1$, (b) $H = 0.5$, (c) $H = 0.9$. The red bars "Exact" indicate the exact distribution of 5000 FGN series in each group. The line "Normal" corresponds to the normal distribution with the mean from Equation (3) and the variance from Equation (4). The line "Norm*" indicates the normal distribution with the mean and variance calculated from the FGN series in "Exact."

hypothesis was mostly rejected when $H \leq 0.5$, while it performed better only when H tended to 0.9. Even in these cases, the GN distribution fitted better than the beta distribution for more groups. Therefore, the GN distribution is considered the better method for fitting the distribution of statistic S .

The parameter k depends on H and N , which equals 2 when $H = 0.5$. The results of the parameter estimation of the same group of series, as listed in Table 1, were taken as the sample. The empirical expression of k on H and N is given as

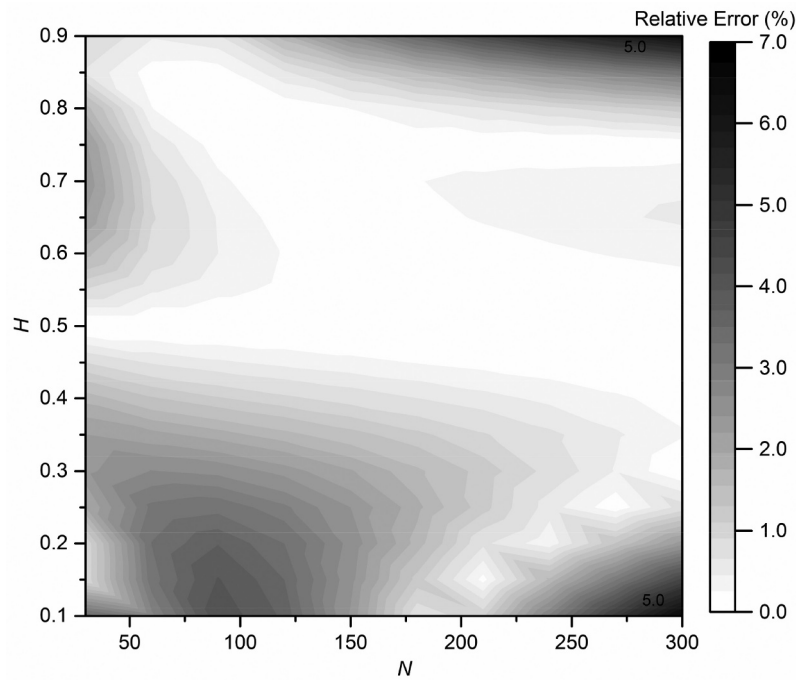


Figure 5. Absolute values of the relative fit error distribution of the variance of S versus persistence level H and series length N .

$$k = 2 + (1.8H - 0.9) \times (0.98 + 0.02H)^N \quad (22)$$

Thus far, both the theoretical and empirical values are given for the S variance of scaling data, and the GN distribution is used to approximate the real distribution of S , instead of the normal distribution. The modified MK trend test is called the MKG in the rest of the paper.

5 Comparison between alternative trend tests

5.1 Evaluation criterion for comparison

The errors in significance tests can be divided into two types. Type I error represents the false rejection of the null hypothesis

H_0 while it should have been accepted. The probability of Type I error α_0 equals the preassigned significance level α ideally while an error always occurs in practice since the actual distribution of the statistic cannot be fitted error-free. Hence, α_0 is used as the main indicator of the goodness of fitting the distribution of S under H_0 in this study.

Type II error represents the false acceptance of H_0 while the series exhibits a trend. The probability of Type II error is represented by β . The power of a test method is defined as $1 - \beta$. With the existence of a fixed trend component, a higher power represents higher sensitivity of the test, which is mainly decided by the definition of the statistic and also influenced by other factors such as the magnitude of the target feature and

Table 1. K-S test statistic results of the distribution fit of the S values of 5000 time series iterations for each group with different N and H . The maximum-likelihood estimation (MLE) is applied for the distribution fit with both the GN and beta distributions. * denotes the rejection of the null hypothesis at the 0.1 significance level.

N	Distribution	H								
		0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
30	GN	0.018*	0.021*	0.015	0.013	0.014	0.011	0.011	0.017	0.012
	Beta	0.019*	0.024*	0.026*	0.020*	0.017	0.012	0.014	0.015	0.009
60	GN	0.013	0.010	0.012	0.015	0.008	0.012	0.015	0.008	0.009
	Beta	0.020*	0.020*	0.021*	0.018*	0.021*	0.015	0.018*	0.010	0.010
90	GN	0.012	0.016	0.139	0.013	0.007	0.008	0.010	0.007	0.009
	Beta	0.028*	0.025*	0.018*	0.018*	0.016	0.014	0.013	0.009	0.007
120	GN	0.010	0.012	0.010	0.007	0.008	0.006	0.011	0.013	0.007
	Beta	0.017	0.021*	0.020*	0.014	0.019*	0.015	0.016	0.016	0.007
150	GN	0.009	0.013	0.005	0.007	0.007	0.009	0.006	0.012	0.010
	Beta	0.017	0.026*	0.014	0.018*	0.018*	0.013	0.009	0.014	0.009
180	GN	0.010	0.009	0.013	0.009	0.010	0.010	0.009	0.009	0.008
	Beta	0.014	0.016	0.015	0.018*	0.014	0.016	0.018*	0.012	0.009
210	GN	0.015	0.009	0.008	0.009	0.009	0.008	0.011	0.012	0.010
	Beta	0.021*	0.016	0.016	0.018*	0.020*	0.014	0.014	0.018*	0.007
240	GN	0.009	0.007	0.008	0.013	0.010	0.008	0.007	0.008	0.009
	Beta	0.021*	0.017	0.015	0.018*	0.012	0.015	0.016	0.014	0.011
270	GN	0.006	0.007	0.012	0.007	0.008	0.010	0.010	0.007	0.008
	Beta	0.017	0.017	0.014	0.015	0.015	0.017	0.019*	0.008	0.009
300	GN	0.006	0.016	0.007	0.010	0.008	0.007	0.008	0.011	0.010
	Beta	0.016	0.019	0.017	0.018*	0.018*	0.017	0.0144	0.0148	0.009

series length. Therefore, the power is also taken as a key criterion for comparing the goodness of the methods.

5.2 Comparison between modifications of S distribution for MK test

The modification effects of the S distribution obtained using the MKG, MKB (the beta distribution with the modified variance), and MKN (the normal distribution with the modified variance) were compared. The MKO was also considered as a contrast. Type I error was tested by conducting 5000 iterations of the Monte Carlo simulations on the stationary series for each group with $H = 0.1 : 0.05 : 0.9$ and $N = 30, 300$ (representing small and large sample sizes, respectively). Figure 6 shows the results. α_0 of MKO increases sharply with increasing H , far beyond 0.05. With larger sample sizes, it increases more rapidly. With the modified variance, the MKG, MKB, and MKN all perform well (α_0 is near 0.05) in the entire range of H with both small and large samples. Moreover, the series length has little influence on α_0 for the three methods, demonstrating the reasonability of the modifications.

Among the three methods, the MKB seems to deviate more from 0.05 with negative persistence; however, this is not evident in Fig. 6. For a further analysis, the tails of the cumulative distributions of the normal, beta, and GN distributions were compared with the cumulative frequency of the samples of $N = 30$. Figure 7(a) shows the tails of $H = 0.1$. The blue line (GN) fits best with the exact cumulative frequency data points, while the red (normal) and green (beta) lines coincide and cross the blue one near the 0.95 cumulative frequency. Figure 7(b) shows the case when $H = 0.9$. The beta distribution fits best in the range of 0.95–0.98, while both the GN and normal distributions give reasonable approximations. With the cumulative frequency above 0.98, the GN distribution fits better than the beta distribution, and the deviation in the normal distribution increases. The best distribution pattern changes between the GN and beta with the variation in the cumulative frequency. However, considering the significance levels typically used in practice (two-sided test with a 95% significance level at least), the GN distribution gives a better approximation of S .

5.3 Comparison between MKG and other trend detection methods

Since the MKG was proven to better fit the S distribution under H_0 than the MKN and MKB, its test ability was further compared with those of the trend-free pre-whitening MK (MKPW, Yue and Wang 2002, Rivard and Vigneault 2009) test, over-whitening MK (MKOW, Sen 2016) test, and regression analysis (RA, Matalas and Sankarasubramanian 2003) in this section. The Monte Carlo simulation was set to 5000 times for each group with $H = 0.1 : 0.05 : 0.9$, $\beta = 0 : 0.005 : 0.05$, and $N = 60, 300$ (representing small and large sample sizes). Figure 8 shows the results of the stationary time series, namely α_0 obtained using the four methods. According to the results, α_0 of the MKG is close to α across the entire range of H with both small and large series lengths. RA also performs well with long series, but α_0 increases to more than 0.1 when $H = 0.9$ with $N = 60$. While the impact of LTP on the variance of the statistic was considered in the RA method, the increase in the probability of Type I error might be due to the lack of consideration of the change in the distribution shape, which is illustrated in Section 3.2, although the statistics are different in the two tests. For both short and long series, α_0 of MKOW is close to α , showing the effectiveness of the OW process in reducing serial correlation. α_0 of MKPW increases sharply with increasing H for both sizes of the time series. This indicates that the pre-whitening process aiming at lag-1 independence cannot effectively eliminate the correlation of LTP data with larger lag, leading to more fake trends than α . In summary, the MKG and MKOW outperform the MKPW for stationary data, since their Type I error probability values are close to α in the entire range of H , regardless of the sample size. Although RA is credible in most cases, it shows some weakness in the case of limited length and strong LTP.

Figure 9 shows the detection rates of nonstationary data, representing the power of the methods. All the methods perform better with long series than with short ones, as the random component in the time series accounts for a larger proportion with smaller sizes. For the same reason, data with larger mean slope have greater probability to be detected with significant trend. Besides the common features, the results of the methods also show a significant difference. 1) MKG: The

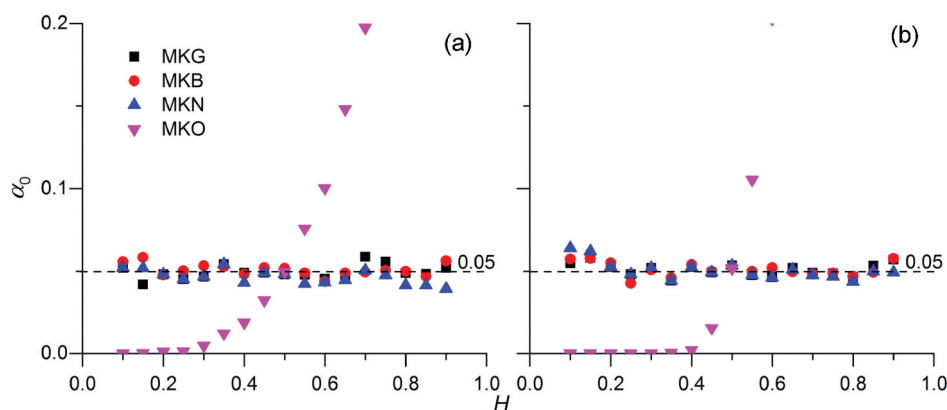


Figure 6. Comparison of Type I error between MKG, MKB, MKN, and MKO methods with (a) $N = 30$, (b) $N = 300$.

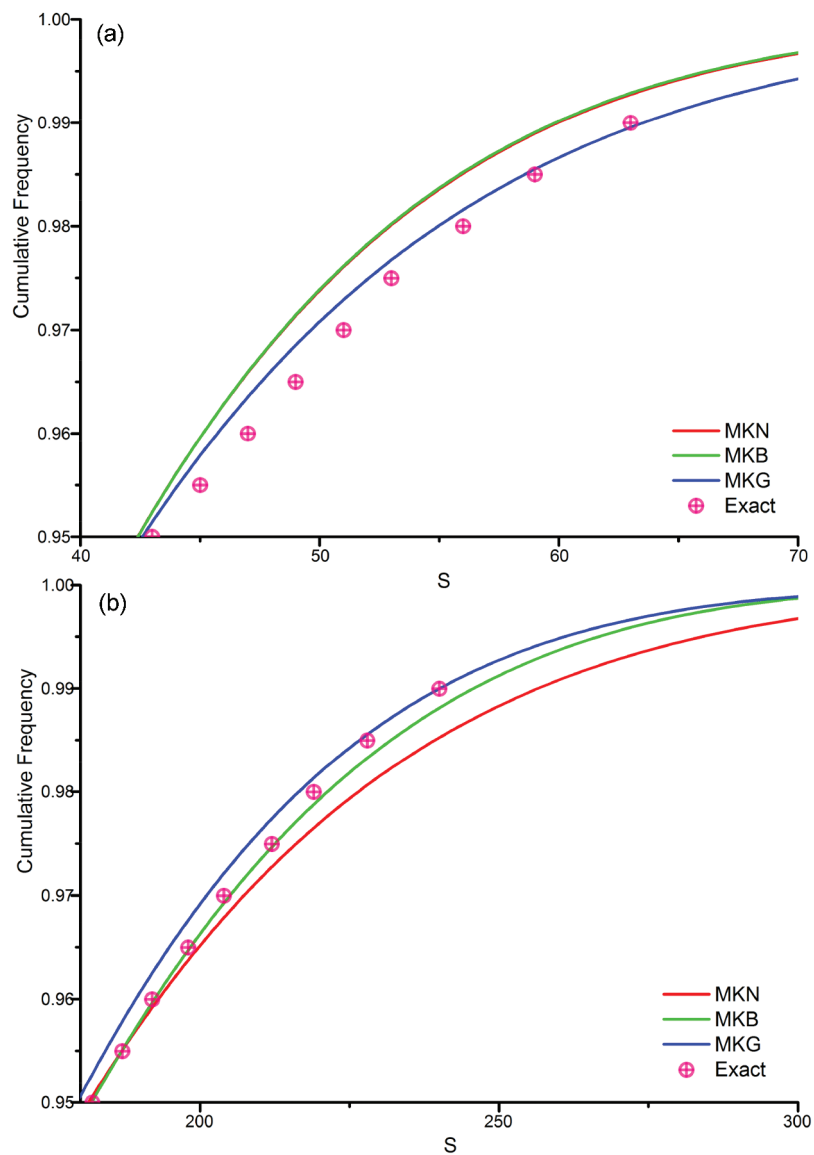


Figure 7. Tails of cumulative distributions of the normal (red), beta (green), and GN (blue) distributions and the exact cumulative frequency of S (the pink points) with $N = 30$ and (a) $H = 0.1$, (b) $H = 0.9$.

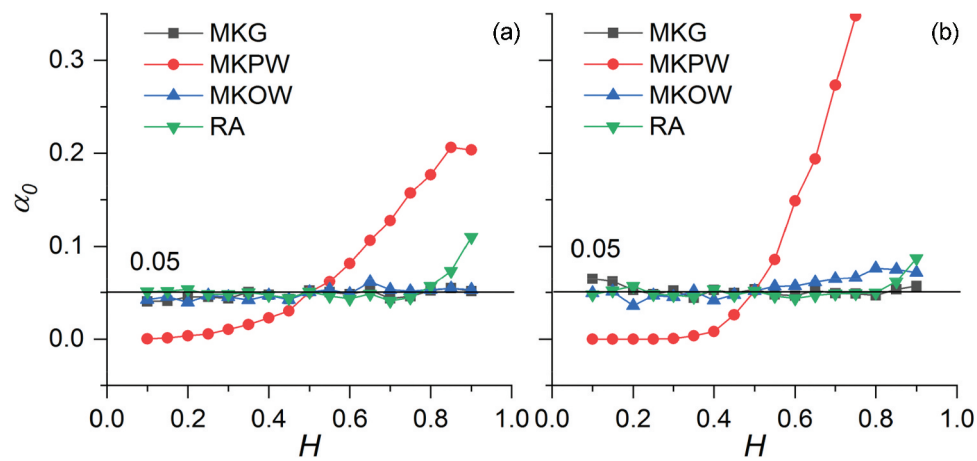


Figure 8. Comparison of Type I error between MKG, MKPW, MKOW, and RA methods with (a) $N = 60$ and (b) $N = 300$ with the preassigned persistence.

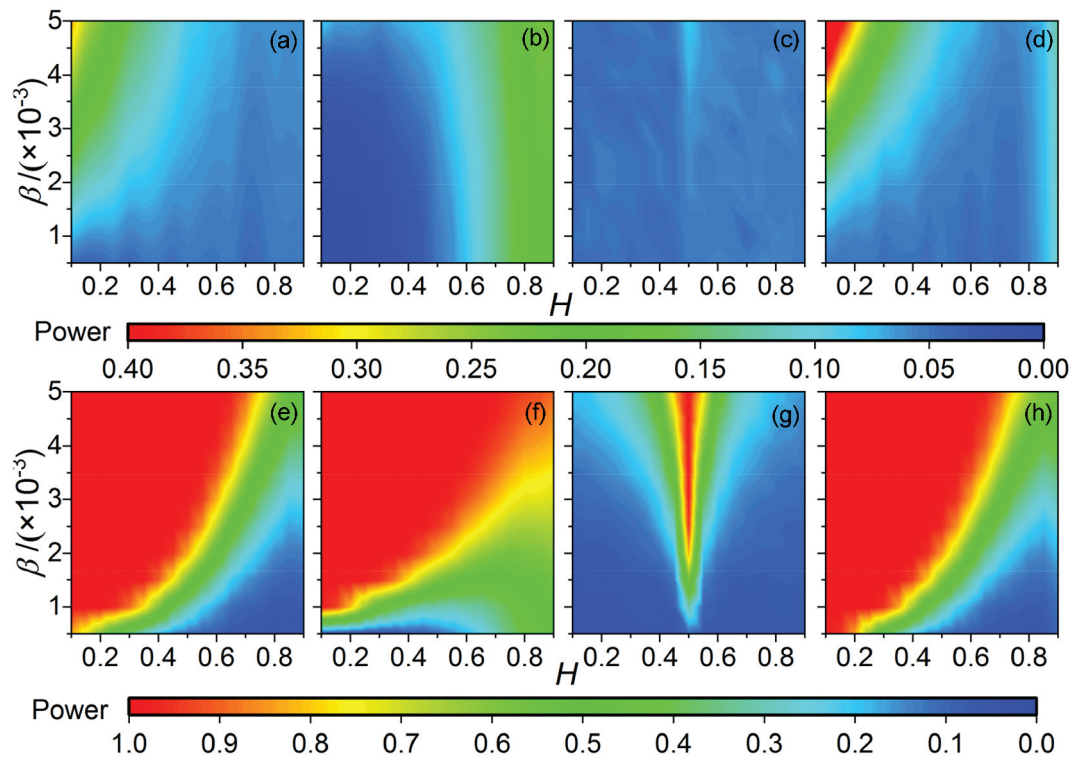


Figure 9. Comparison of power between (a) MKG with $N = 60$, (b) MKPW with $N = 60$, (c) MKOW with $N = 60$, (d) RA with $N = 60$, (e) MKG with $N = 300$, (f) MKPW with $N = 300$, (g) MKOW with $N = 300$ and (h) RA with $N = 300$ with preassigned persistence.

power with the same series length increases with decreasing H . This is because of the changes in the variance and distribution shape with the variation in the persistence level. The stronger the negative persistence of the data, the narrower the distribution of S . Therefore, there is a greater probability for the same trend component to fall outside the confidence interval. On the contrary, S of stronger positive persistence is distributed more widely, thus reducing the probability of trend detection. 2) MKPW: The performance is similar to that of MKG in general; however, the change law of the power versus H is not constant, as it depends on the residual serial correlation after pre-whitening process. Moreover, it has greater power than MKG with long time series. 3) MKOW: The power is lower than those of MKG and MKPW with nonstationary data. For long series, the power is lower with both positive and negative persistence than independent data, owing to the added white noise series. As mentioned in Section 1, although a modification can reduce serial correlation while retaining the trend component, it weakens the significance of the trend when adding additional randomness. 4) RA: The performance is quite similar to that of MKG, except that it is more powerful than MKG for small sample size and strong negative LTP. Such a difference might be due to its parametric feature, making it usually more powerful than non-parametric tests because the data themselves are tested instead of their ranks. However, for the same reason, parametric tests are sensitive to the distribution of the variable and are influenced by skewness and outliers.

The newly developed innovative trend analysis (ITA, Sen 2012) is worth discussing. As stated by Sen, the method is free from any assumptions and is not influenced by serial

correlation and limited length. Since ITA relies on subjective judgement after the half-half relationship is plotted, it cannot be compared with the MKG by Monte Carlo simulations. Figure 10 shows the half-half plots of four time series with $N = 60, 300$ and lag-1 correlation $\rho_1 = 0, 0.8$, respectively representing short and long series and independent and dependent data. The results visually show some drawbacks of the ITA method. The data points in the red circles, in Fig. 10(a) and (c), deviate from the 45° line while the time series is stationary in reality. With respect to the ITA, an increasing trend in the high values and a decreasing trend in the low values are observed, while they actually appear random due to the limited sample size. The misestimation is due to the lack of consideration of the randomness in the time series, which is particularly evident in extreme ranges. To be specific, the horizontal coordinate x_i and vertical coordinate y_i of any data point in the half-half plot are the values of the same rank in the two half-series, which contain randomness being observations of a stochastic variable. If the series is long enough, the frequency distribution of both the half-series are close to the probability distribution of the variable, hence $x_i \approx y_i$ under the stationarity assumption. With the limited sample size of hydrological observations, the error between the frequency distributions of the two half-series might be nonnegligible, thus deviating (x_i, y_i) from the 1:1 line. The deviation is especially large in the extreme ranges, as the tails of the variable distribution are usually thinner than the middle parts. The ITA requires a large series length to ensure that the influence of serial correlation is eliminated (Fig. 10(b)). With the limited length, a stationary time series is likely to show a fake trend in the ITA plot (the red line in Fig. 10(d)).

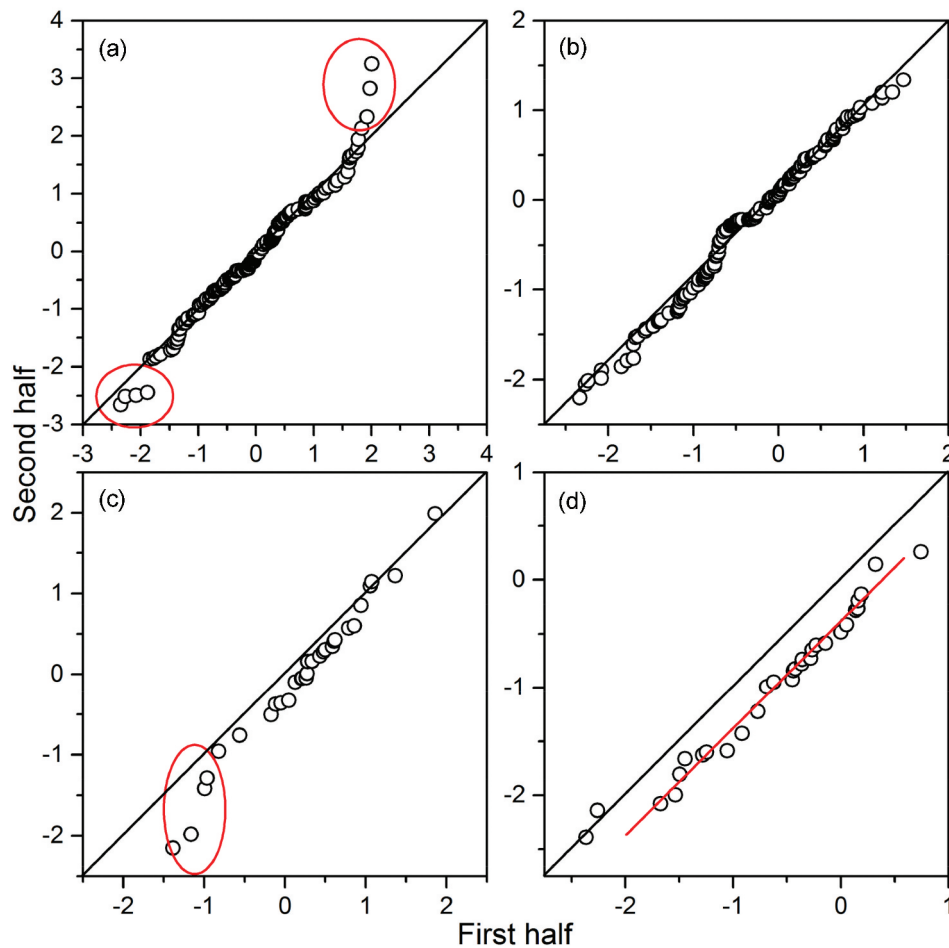


Figure 10. Half-half plots for innovative trend analysis (ITA) of stationary time series with (a) $N = 300, \rho_1 = 0$, (b) $N = 300, \rho_1 = 0.8$, (c) $N = 60, \rho_1 = 0$, (d) $N = 60, \rho_1 = 0.8$.

Moreover, the ITA relies on human judgement and lack numerical criterion, making it inefficient when dealing with mass data.

Overall, among the MKG, MKPW, MKOW, and RA, the MKG and MKOW perform better with stationary data, with α closer to α_0 , while the MKG, MKPW, and RA show greater power with nonstationary data. The newly developed ITA method shows some drawbacks because of the lack of consideration of randomness, with poor performance in extreme ranges and with limited series length. Therefore, the MKG is demonstrated to be the best trend significance test among the alternatives.

6 Estimation of H under the scaling hypothesis

6.1 Comparison of alternative estimators of H

The modification of the S distribution is based on a preassigned value of H , which is unknown in practice. Therefore, some estimators must be applied to obtain the H value. To this end, four typical methods were chosen as alternatives: Rescaled Range (R/S analysis, Mandelbrot and Wallis 1968), Higuchi method (Higuchi 1988), Periodogram (Taqqu *et al.* 1995), and CDMA (Arianos and Carbone 2007).

Five thousand Monte Carlo simulations were performed for each group with $\beta = 0$ (0.001) 0.009, $H = 0.1$ (0.1) 0.9, and $N = 2^5, 2^6, \dots, 2^{10}$ to determine the best approach. When synthetic data are stationary, all the four methods perform well for large sample sizes. With decreasing size, taking $N = 64$ as an example (shown in Fig. 11), the periodogram method performs poorly because of the unacceptable bias and randomness. Among the other three methods, the CDMA gives the best performance in terms of the bias and uncertainty. The estimation is unbiased near $H = 0.5$ while underestimated (overestimated) when $H > 0.5$ ($H < 0.5$). The Higuchi method exhibits less bias than the R/S analysis; however, both overestimate in most of the range of H while underestimating for large H values. The uncertainty of H estimations by the R/S analysis is less than that by the Higuchi method, particularly for small H values.

The existence of trends may contaminate the results of the Higuchi method and R/S analysis (Yue and Wang 2004, Hamed 2008). Hamed and Rao (1998) applied the Sen's trend test (Sen 1968) and eliminated the apparent trend from the original time series before estimating H (denoted as Sen's detrending). This pretreatment method has been widely used in significance tests on trends. However, there remain some doubts for the following reasons. 1) Compared with adjustment of the distribution of the

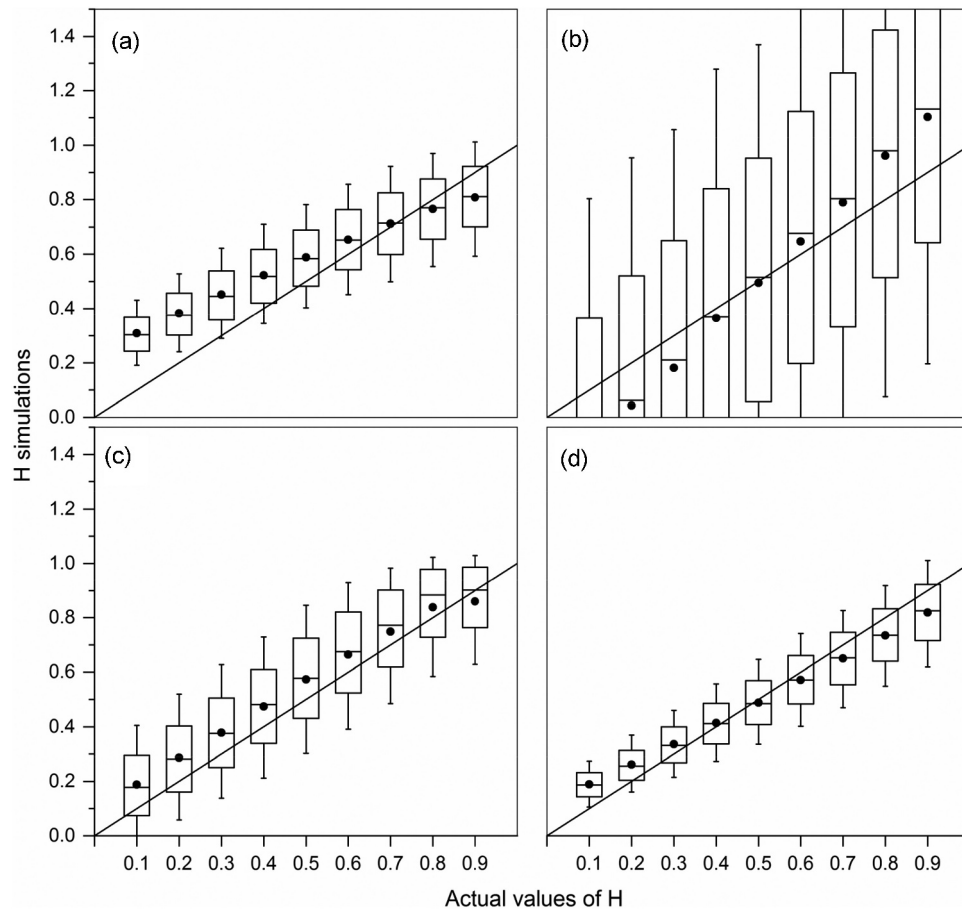


Figure 11. H values estimated by (a) R/S analysis, (b) Periodogram, (c) Higuchi method, and (d) CDMA method of FGN series with $N = 64$ with the absence of trend component. The range of the boxes is 25–75%, the whiskers indicate the 10 and 90% values, and the circles indicate the average values.

statistic, modifying the original time series seems inappropriate for dealing with trend components, as it could cause loss of information. 2) The trend components in hydrological data are not always linear as assumed in

Sen's detrending method: it could be quadratic, polynomial, or any other complex pattern. It is difficult to find an appropriate function to describe and eliminate the trend components.

Table 2. Mean and variance of the estimation results of the H value with CDMA and SH methods. The series used are generated from the normal distribution $N(0, 1)$, with different patterns and magnitudes of the slopes added to the mean or variance.

		1	2	3	4	5	6	7	8	9
Linear Mean $N = 500$		Slope of the trend ($\times 10^{-3}$)								
CDMA	Mean	0.517	0.515	0.517	0.519	0.518	0.518	0.521	0.522	0.522
	Variance($\times 10^{-3}$)	1.250	1.133	1.237	1.177	1.156	1.146	1.229	1.155	1.147
SH	Mean	0.526	0.530	0.530	0.527	0.526	0.534	0.522	0.525	0.526
	Variance($\times 10^{-3}$)	12.589	12.717	12.332	12.778	12.465	12.198	12.556	12.368	11.843
Linear Mean $N = 50$		Slope of the trend ($\times 10^{-3}$)								
CDMA	Mean	0.525	0.524	0.518	0.528	0.524	0.522	0.527	0.529	0.521
	Variance($\times 10^{-3}$)	11.063	12.328	11.024	11.000	11.164	10.976	10.642	12.124	11.582
SH	Mean	0.543	0.542	0.548	0.546	0.547	0.545	0.547	0.544	0.551
	Variance($\times 10^{-3}$)	31.720	32.489	31.336	31.895	32.502	31.219	32.104	31.603	31.775
Linear STD $N = 500$		Slope of the trend ($\times 10^{-3}$)								
CDMA	Mean	0.511	0.510	0.508	0.506	0.504	0.503	0.498	0.499	0.501
	Variance($\times 10^{-3}$)	1.255	1.318	1.475	1.483	1.412	1.711	1.703	1.529	1.786
SH	Mean	0.523	0.501	0.503	0.501	0.499	0.501	0.500	0.498	0.495
	Variance($\times 10^{-3}$)	10.920	10.702	10.323	10.617	10.579	10.626	11.370	11.332	10.822
Quadratic Mean $N = 500$		Slope of the trend ($\times 10^{-5}$)								
CDMA	Mean	0.530	0.532	0.534	0.536	0.537	0.543	0.546	0.550	0.559
	Variance($\times 10^{-3}$)	1.014	0.957	1.011	1.058	1.014	0.866	0.945	0.085	0.873
SH	Mean	0.956	1.026	1.038	1.041	1.043	1.044	1.044	1.044	1.044
	Variance($\times 10^{-3}$)	2.517	0.079	0.019	0.010	1.008	0.007	0.006	0.004	0.004

The Higuchi method was taken as the representative for illustrating the drawbacks of Sen's detrending pretreatment. To compare CDMA and Sen's detrending Higuchi (SH) analysis, 5000 synthetic series were generated for each group with different trend patterns and series lengths. Table 2 lists the results. The linear trend in the standard deviation seems to have no evident influence on the unbiasedness of the two estimates; the results of SH show more uncertainty. The SH analysis performs well with linear mean trends when the series is long enough ($N = 500$) because the detrending step eliminates the linear trends estimated by Sen's slope, of which the impact of randomness is negligible due to the large sample size. With smaller size ($N = 50$), the randomness affected the SH results with an unacceptable variance. When the trend component turned to quadratic, the SH failed to give accurate estimates because of the shortcoming of the detrending step as stated before. In contrast, the CDMA performs throughout. Hence, it is regarded as the best approach for LTP estimation of hydrological time series.

6.2 Centred detrending moving average algorithm

For a time series $x(t)$, $t = 1, 2, \dots, N$, the cumulative sum sequence can be calculated by

$$X(t) = \sum_{i=1}^t x(i), \quad t = 1, 2, \dots, N \quad (23)$$

We consider a moving window $[t - s_1, t + s_2]$ with the size of s , where $s_1 = (s - 1)/2$, $s_2 = (s - 1)/2$, $\lfloor \cdot \rfloor$ means rounding down, and $\lceil \cdot \rceil$ means rounding up. The moving average $\tilde{X}(t)$ is given by

$$\tilde{X}(t) = \frac{1}{s} \sum_{k=-s_2}^{s_1} X(t - k) \quad (24)$$

By eliminating the moving average $\tilde{X}(t)$ from the cumulative sums $X(t)$, we can express the detrended residual as

$$\epsilon(t) = X(t) - \tilde{X}(t), \quad s_1 \leq t \leq N - s_2 \quad (25)$$

We divide into N_s disjoint blocks with the length of s , where $N_s = \lfloor \frac{N}{s} \rfloor - 1$. The root-mean-square $F(s)$ with the window size of s is given by

$$F^2(s) = \frac{1}{sN_s} \sum_{v=1}^{N_s} \sum_{i=1}^s \epsilon^2[(v-1)s + i] \quad (26)$$

$F(s)$ presents a power-law relation with s , expressed as

$$F(s) \sim bs^H \quad (27)$$

$$\ln[F(s)] \sim \ln(b) + H \ln(s) \quad (28)$$

where b is a constant. Thus, the slope of $F(s)$ - s on the log-log plot is regarded as the H value of the series.

According to Shao *et al.* (2015), the slope of the log-log plot represents long-term persistence with low values of s , while it represents the trend with large s . The critical value s_x is determined by H and β and also by the trend model chosen. For a linear trend, s_x equals $\left(\frac{24b}{\beta}\right)^{1/(2-H)}$. With the parameters

correctly set, the CDMA analysis is adequate for H estimation of series with trend components.

6.3 Impact of the uncertainty in H estimation on MKG

As presented in Section 6.1, although the CDMA performs the best among the alternative H estimators, its estimation still shows bias and uncertainty when the series length is limited. The bias is due to the failure of meeting the assumption of infinite length when the time series is not long enough, thus increasing the error of some approximations in the estimators. The uncertainty in the estimations can be attributed to the randomness of the stochastic process. In other words, a time series is a finite-size sample of a stochastic variable, containing randomness itself. The shorter the time series, the more significant the randomness. Therefore, the unbiasedness might be improved by developing estimators that are less dependent on the infinite-length assumption, while the uncertainty can only be reduced and not completely eliminated.

The MKG is the best trend test method among the alternatives in cases wherein the H values are already known, as demonstrated in Section 5.3. However, the bias and uncertainty in H estimation would contaminate the test results of the MKG. Monte Carlo simulations were performed to quantitatively evaluate the influence. The parameter set for the statistical experiments was the same as the one given in Section 5.3: $H = 0.1 : 0.05 : 0.9$, $\beta = 0 : 0.005 : 0.05$, and $N = 60, 300$. $N = 60$ was set for the representative of the common length available in present hydrological records, while $N = 300$ was set to obtain the performance of the methods as the sample size grows. The MKG test was applied with both preassigned H values (denoted as MKG-Known) and H estimated by CDMA (denoted as MKG-CDMA). Additionally, the Higuchi method and R/S analysis were applied for comparison when testing Type I error, denoted as MKG-Higuchi and MKG-R/S, respectively.

Figure 12(a) shows the probability of Type I error of MKG with $N = 300$. The MKG-Known and MKG-CDMA both show high accuracy, as α_0 is close to the preassigned value of 0.05 throughout. The result of the Higuchi method is also good in most cases, except that α_0 exceeds 0.1 when $H = 0.9$. The MKG-R/S shows the worst result among the four tests, as α_0 is close to α only with $H = 0.5-0.8$. When the series gets shorter (Fig. 12(b)), MKG-Known maintains its high accuracy in the entire range of H , indicating the applicability of MKG test to short series. For MKG-CDMA, α_0 increases from 0.05 to 0.117 as H increases from 0.1 to 0.9. As stated, the error comes from the uncertainty during the estimation of H and could be reduced by lengthening the records or developing better algorithms. Before that, the performance of MKG-CDMA with moderate-length records is considered to be acceptable in practice. The performances of MKG-Higuchi and MKG-R/S are quite similar when $H = 300$, suggesting that they are less affected by the variation in the series length. In summary, CDMA performs better than the R/S analysis and Higuchi method along with MKG test since it exhibits the least influence on the Type I Error of MKG. Moreover, it is demonstrated that the performance of MKG-CDMA will improve with record length increasing. Therefore, CDMA is recommended as the H estimator before MKG test.

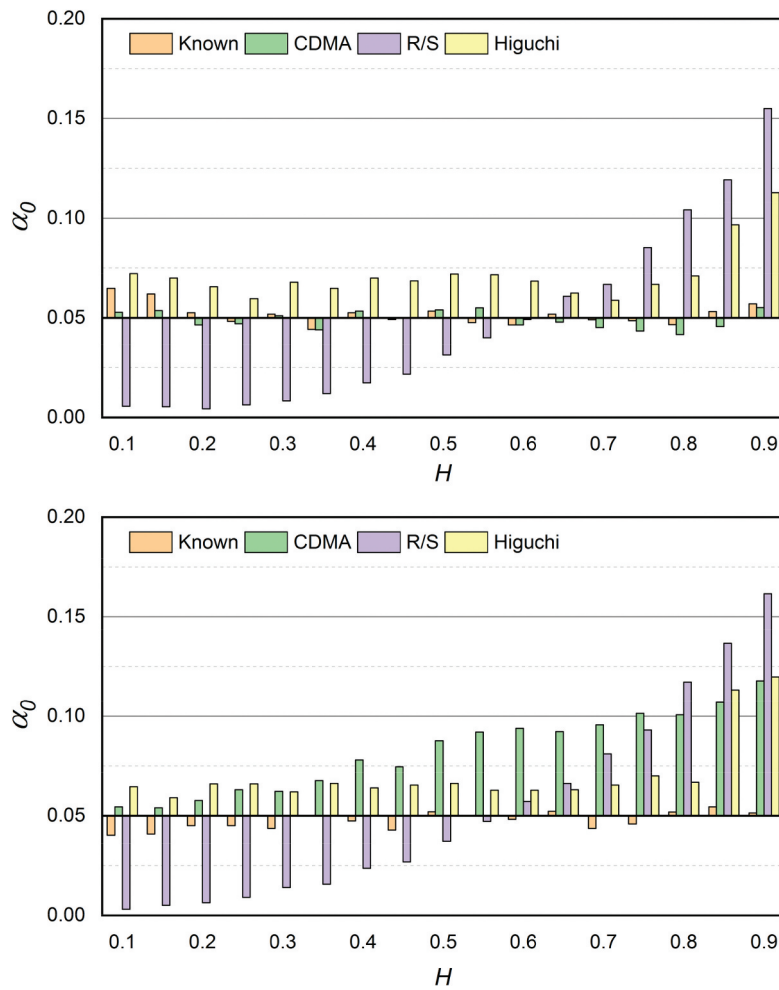


Figure 12. Comparison of Type I error of MKG with known H and H values estimated by CDMA, R/S analysis, and Higuchi method with (a) $N = 300$ and (b) $N = 60$.

Figure 13 shows the influence of H estimation by CDMA on the Type II error of MKG. When the time series is long (Fig. 13(c) and (d)), the power values of MKG-Known and MKG-CDMA seem to be quite similar. However, when the series length decreases (Fig. 13(a) and (b)), MKG-CDMA shows a slightly lower detection rate than MKG-Known under the same H and β . This indicates that the bias and uncertainty in H estimation by CDMA affect the results of MKG, particularly in the case of shorter time series. However, the influence is still acceptable at a series length of 60, which is typical in hydrological observations.

In summary, there is bias and uncertainty in the H estimation of finite-length series, owing to the limited series length of the stochastic process. With the CDMA, the influence on MKG can be drastically reduced to an acceptable level compared with the widely used R/S analysis. In future studies, new algorithms should be developed to further reduce the bias and uncertainty in H estimation. More importantly, the uncertainty that cannot be eliminated should be reflected in the result of the estimation to guide subsequent analyses.⁷ Application to non-Gaussian data.

7 Application to non-Gaussian data

For independent data, the MKO is free from the distribution of the observation values. However, when taking persistence into

account, as explained in Section 2, the analyzed data are assumed to be normally distributed. The H estimators were compared for the Gaussian samples in Section 6.1. To keep the distribution-free characteristic of the MK test, normalization pretreatment is required for raw data. The observations can be transformed into an equivalent Gaussian series $Y = \{y_1, y_2, y_3, \dots, y_N\}$ by

$$y_i = \Phi^{-1}\left(\frac{R_i}{N+1}\right) \quad (29)$$

where R_i is the rank of x_i , and $\Phi^{-1}(x)$ is the inverse function of CDF of the standard normal distribution. Because the series Y is generated based on the ranks of series X , the corresponding rank sequences of the two time series are the same. Therefore, X and Y must have the same significance for the statistic S in the MK test, since the MK test is essentially a test for the rank sequence. Therefore, Y is proposed to be the substitute of X for the MK test. Both H estimation and trend detection should be applied to Y instead of X .

Based on the former discussion, the procedure of MKG for scaling data is summarized as follows:

Step 1: Calculate the value of statistic S of the target time series using Equation (1).

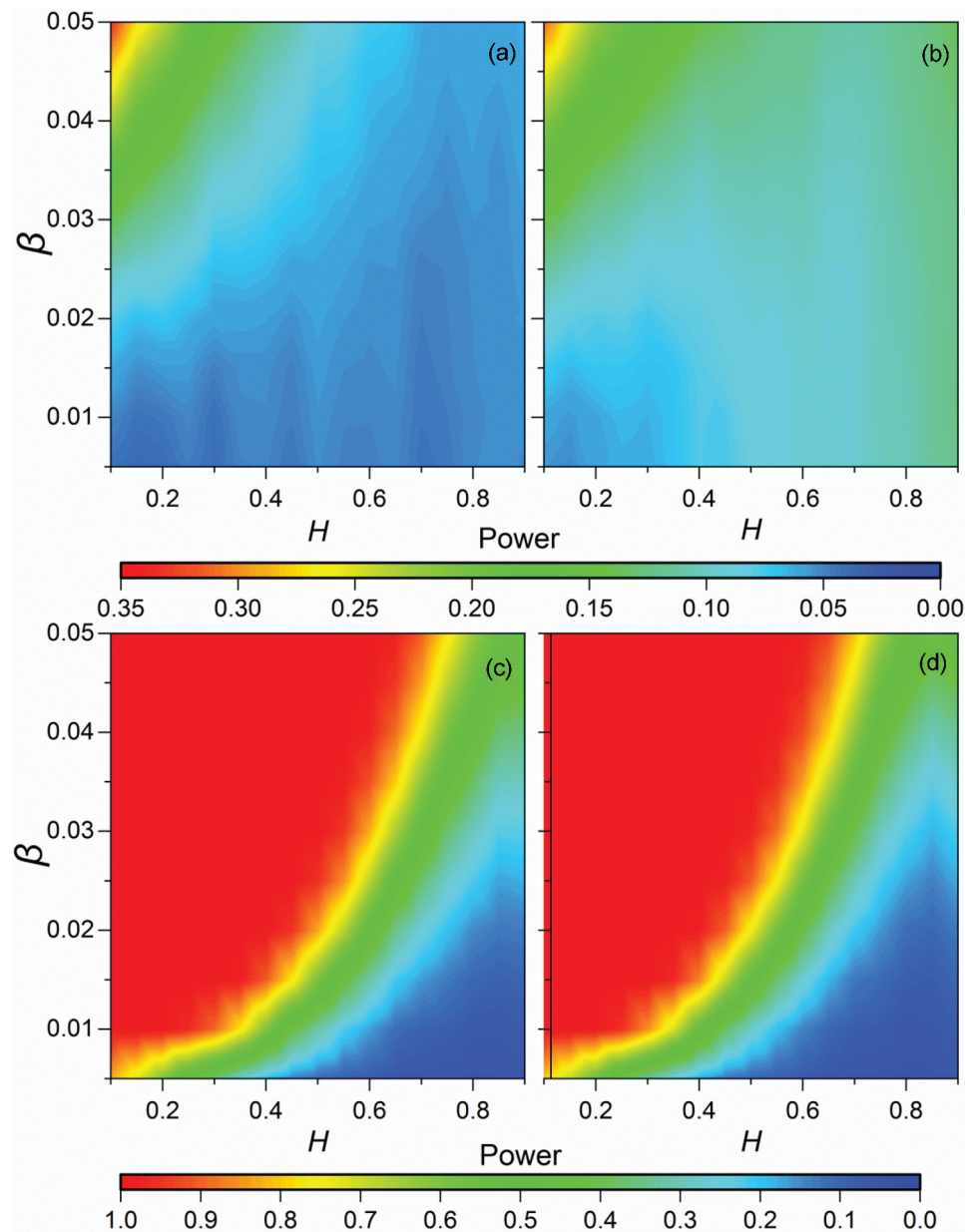


Figure 13. Comparison of power between MKG with (a) $N = 60$ and known H , (b) $N = 60$ and estimated H , (c) $N = 300$ and known H and (d) $N = 300$ and estimated H .

Step 2: Obtain the equivalent standard normal series Y using Equation (29).

Step 3: Apply Equation (29) to Y for estimating the scaling coefficient H .

Step 4: Calculate the variance of S of given N and H using either theoretical method (Equations (12) and (16)) or empirical method (Equations (18) and (19)).

Step 5: Calculate the shape parameter k using Equation (22)

Step 6: Comparing S with the critical value of the specific significance level, judge the trend significance.

8 Case study

8.1 Study area and data

The MKG procedure was applied to the annual total precipitation data obtained from nine hydrometric stations located in

China. The stations are located in three provinces, namely Xinjiang, Jiangsu, and Guangxi, as shown in Fig. 14. The daily total precipitation data were provided by the China Meteorological Data Service Centre, China Meteorological Administration ([http://data.cma.cn/data/cdcdetail/dataCode/](http://data.cma.cn/data/cdcdetail/dataCode/SURF_CLI_CHN_MUL_)

DAY_V3.0.html). Table 3 lists the location of the stations and the statistical characteristics of the data. The series length is in the range of 58–66 years with the absence of missing data, which meets the requirements of hydrological statistical tests and can reflect the impact of finite series length. Xinjiang is located in the northwest frontier of China. Far inland from the sea with surrounding mountains, Xinjiang shows a typical temperate continental climate with low rainfall. The annual average precipitation (AAP) is only approximately 150 mm, the interannual variation is large, and data

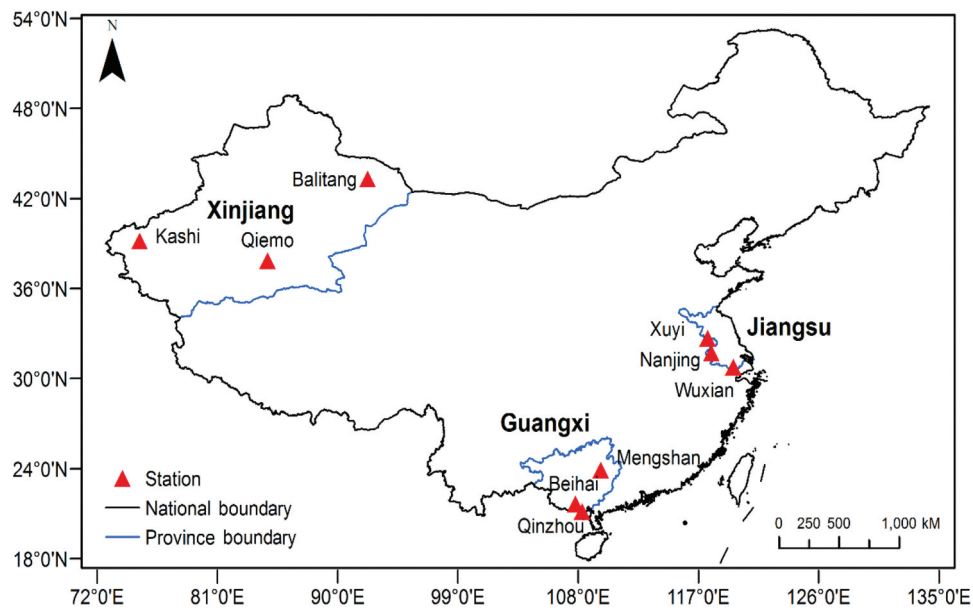


Figure 14. Locations of the nine hydrometric stations in China applied in the study.

Table 3. Details of the stations and dataset used in the case study and their statistical characteristics. AAP represents the average annual precipitation of the time series, and the parameters C_v and C_s denote the coefficient of variation and coefficient of skew, respectively.

Province	Station	Station ID	Latitude	Longitude	Time period	AAP (mm)	C_v	C_s
Xinjiang	Kashi	51709	39.47 N	75.98 E	1954–2017	71.37	0.533	0.997
	Qiemo	51855	38.15 N	85.55 E	1959–2017	22.98	0.554	0.369
	Balitang	52101	43.60 N	93.05 E	1957–2017	224.30	0.222	0.138
Jiangsu	Xuyi	58138	32.98 N	118.52 E	1957–2017	1026.71	0.237	0.254
	Nanjing	58238	32.00 N	118.80 E	1951–2017	1081.09	0.243	0.829
	Wuxian	58358	31.07 N	120.43 E	1956–2017	1156.81	0.180	0.388
Guangxi	Mengshan	59058	24.20 N	110.52 E	1955–2017	1757.10	0.182	0.226
	Qinzhou	59632	21.95 N	108.62 E	1953–2017	2158.35	0.169	−0.224
	Beihai	59644	21.45 N	109.13 E	1954–2017	1756.60	0.221	0.351

from stations (e.g. Kashi station) exhibit significant skewness. Jiangsu is located on the east coast of mainland China in the Yangtze River Delta with a dense river network and lakes. Located in the transition zone of the subtropical and warm temperate zones, the climate has both southern and northern characteristics. The AAP in Jiangsu is approximately 1000 mm with little spatial and interannual variation. Some stations here (e.g. Nanjing station) also exhibit strong skewness. Guangxi is located on the south coast of mainland China with a typical subtropical monsoon climate with good rainfall. The AAP of this region is up to 1700 mm with little temporal variation and skewness.

8.2 Results and discussion

In addition to the MKG method proposed in this study, some other trend tests were applied to the measurements. Linear regression (LR) and Sen's slope estimation were used to estimate the magnitude of the linear slope. Because of the lack of consideration of persistence, the results of the two estimators are simply taken as reference of the apparent trend magnitude. Three other trend significance tests, namely MKPW, MKOW, and MKO, were applied to compare the MKG. Figure 15 shows the annual total precipitation at the nine stations. Table 4 lists the test results.

Kashi station. The data present strong negative persistence. LR and Sen both indicate a significant increasing trend. The slope from LR is larger than the one from Sen, because of the several outliers in the second half of the time series (Fig. 13a). Because of the large magnitude of the apparent trend, the existing persistence does not influence the test results of MKG, MKPW, and MKO (the p-value remains unchanged while the conclusion of significance is the same). Only the MKOW fails to detect the significant trend, because of the additional randomness from the white noise added in the whitening process.

Qiemo station. The time series shows positive persistence. LR and Sen both indicate a significant increasing trend, which is probably the local performance of LTP. Therefore, MKG and MKOW refused to reject the null hypothesis. MKO wrongly rejected the hypothesis because of the lack of consideration of LTP. MKPW detected a fake trend as it eliminates the low-lag correlation and is still influenced by LTP.

Balitang station. The annual precipitation presents a similar apparent trend and weaker negative persistence compared to the data from Kashi station. The p-value of MKOW is lower because the added white noise is with smaller variance. However, it still failed to detect the real trend.

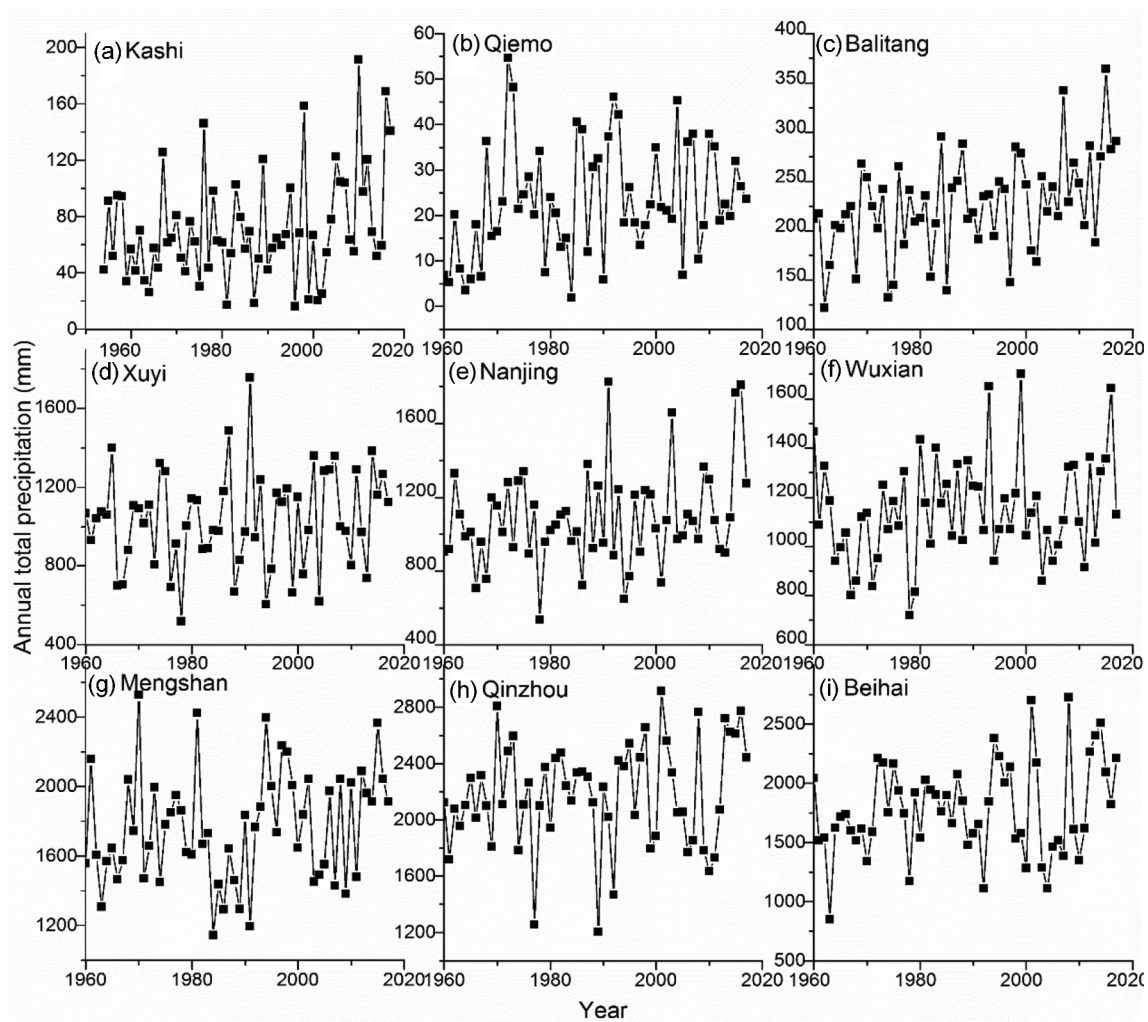


Figure 15. Time series of the annual total precipitation at the nine hydrometric stations in China.

Table 4. Results of the MKG, MKPW, MKOW, and MKO tests. * means the existence of a significant trend at the 0.05 significance level.

Province	Station	H	$LR/\mu (\times 10^{-3})$	$Sen/\mu (\times 10^{-3})$	p-MKG	p-MKPW	p-MKOW	p-MKO
Xinjiang	Kashi	0.220	8.457	5.895	0.001*	0.026*	0.656	0.039*
	Qiemo	0.628	8.911	10.980	0.079	0.044*	0.174	0.017*
	Balitang	0.385	5.752	5.264	0.001*	0.001*	0.282	0.001*
Jiangsu	Xuyi	0.326	2.226	2.735	0.022*	0.078	0.799	0.140
	Nanjing	0.361	2.880	2.105	0.018*	0.058	0.965	0.103
	Wuxian	0.505	1.205	0.875	0.504	0.520	0.520	0.512
Guangxi	Mengshan	0.752	2.133	2.497	0.351	0.270	0.749	0.049*
	Qinzhou	0.463	2.131	2.719	0.023*	0.029*	0.219	0.040*
	Beihai	0.479	3.751	3.851	0.016*	0.019*	0.300	0.023*

Xuyi station. The data exhibit some negative persistence and a slight increasing trend, clearly detected by MKG. As negative LTP would mask the real trend, the MKOW, MKPW, and MKO all failed to detect the trend.

Nanjing station. Owing to geographic proximity, the characteristics of the data from Nanjing station are similar to those from Xuyi station, except for its strong skewness. The test results are also similar, indicating that the methods are free from data distribution.

Wuxian station. The time series is mostly independent, with a slightly increasing apparent trend. The four methods give similar p-values in this case.

Mengshan station. The data show stronger positive LTP than that from Qiemo station, as shown in Fig. 13(g). Both LR and Sen estimated a slight increasing trend, which was judged as “local trend” with MKG, MKPW, and MKOW. Only MKO wrongly regarded it as a significant trend.

Qinzhou station. The annual precipitation presents LTP and apparent trend similar to that in Kashi and Balitang stations, while the annual precipitation is much higher, and the distribution is slightly skewed. The results are also similar to those obtained for the two stations.

Beihai station. The distribution of annual total precipitation presents positive skewness, while the other statistical

characteristics and test results of the station are similar to those of Qinzhou station. It demonstrates the robustness of the skewness of MKG.

All the four significance tests are not influenced by the variation and skewness of the distribution, because of their rank-based characteristics. MKO, MPOW, and MKPW separately presented the same disadvantages, as illustrated in Section 5.3. The MKG modifies the distribution of S instead of changing the data, which ensures the power for data with strong LTP. The method is proven to be reasonably applicable to hydrological time series.

9 Conclusion

Based on the influence of LTP on the MK trend test, an MKG method with modified S variance and GN distribution was developed in this study. The CDMA was recommended for H estimation to reduce the influence of uncertainty. The performance of the MKG was evaluated by comparing it with other widely used trend detection methods for both synthetic and observed data. The following conclusions can be drawn from the results:

(1) Positive LTP increases the variance and widens the distribution, while negative LTP has the opposite effect. Therefore, a “fake trend” in the positive persistent data might be detected, and the real trend in the negative persistent data might be hidden with MKO.

(2) The empirical expression for the S variance proposed in this study can reduce the computational cost while the accuracy remains high, showing strong applicability to hydrological data.

(3) The GN distribution can better fit the S distribution of moderate-length scaling data than the normal and beta distributions in the entire range of H , as demonstrated by performing several thousand iterations of Monte Carlo simulations.

(4) Tested with both synthetic and observed data, the MKG gives less Type I errors than MKPW, MKO, and RA and less Type II errors than MKOW and MKO. Therefore, the MKG is considered the best trend significance test method among the alternatives.

(5) The CDMA method is proven to be the best estimator of H among the alternatives, as it maintains good unbiasedness and low uncertainty. Therefore, it can be applied for H estimation before trend detection.

In the present study, the distribution of the statistic S in the MK test was first adjusted based on the LTP feature for both positive and negative persistent data. The newly developed CDMA method was applied to hydrological observations to ensure the accuracy of H estimation. Hence, the application of the MK trend test can be extended to LTP data without having to adjust the observations to meet the assumption of independence, which is significant given the increasingly prominent nonstationarity of hydrological observations.

However, the uncertainty in H estimation affects the MKG test. Although it decreases to an acceptable level in practice with the CDMA, the uncertainty remains. Therefore, further research is required to develop new algorithms that can reduce the uncertainty in H estimation, and the uncertainty that

cannot be eliminated should be reflected in the estimation result to guide subsequent analyses.

Disclosure statement

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ORCID

Shuguang Liu  <http://orcid.org/0000-0002-8637-4745>

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